

Yanghonghui Chen

Tel: +(86) 13396886608 | Email: cyhh1443560270@gmail.com | Website: <https://yonghonghui.github.io/>

EDUCATION

University of Illinois Urbana-Champaign

Urbana US

Bachelor of Science in Electrical Engineering - GPA: 3.88/4.0

September 2021 – June 2025

Zhejiang University

Haining China

Bachelor of Engineering in Electrical Engineering - GPA: 3.95/4.0

September 2021 – June 2025

Relevant coursework: Data Structures(A-), Analog Digital Signal Processing (A+), Digital Signal Processing(A), Embedded DSP Laboratory (A+), Digital Systems Laboratory(A), IoT and Cognitive Computing (A), Control Systems (A), Machine Learning(B+), Applied Parallel Programming(B+), Semiconductor Electronics(A-), MEMS Devices & Systems(A).

RESEARCH EXPERIENCES

Four-Axis Vacuum Stage for Advanced Nano-Manufacturing

Senior Design (Advisor: Prof. Oleskiy Penkov)

February 2025 - May 2025

Keywords: STM32, RS-485, Stepper Motor Control, PCB Design, HMI, FSM, C/C++

- Control System Design: Engineered an STM32-based control system for a 4-DOF robotic arm, using the RS-485 protocol for robust communication with four stepper motor drivers. Developed control logic for precise multi-axis motion and implemented a Finite State Machine (FSM) to automate complex coating sequences.
- Custom PCB Development: Independently designed and validated a custom PCB to integrate the power distribution (24V) and RS-485 communication bus. The design minimized signal interference and voltage drop, ensuring system reliability in a high-vacuum environment.
- HMI & User Interface: Created an intuitive HMI using a TFT touchscreen and physical buttons. The interface provides real-time monitoring of motor status (speed, position) and allows for one-touch execution of preset programs and emergency stops.

Scalable EEG Signal Classification via Federated Learning

Senior Thesis (Advisor: Prof. Howard Yang)

February 2025 - May 2025

Keywords: TensorFlow, Federated Learning, Personalized FL (pFedMe), EEG, Seizure Prediction

- Designed and implemented a privacy-preserving pipeline for seizure prediction, processing EEG signals into Pearson Correlation Coefficient (PCC) matrices and training a CNN classifier using four distinct paradigms: Local, Centralized, FedAvg, and pFedMe.
- Demonstrated that standard FedAvg underperforms due to client data heterogeneity and severe class imbalance, while personalized pFedMe significantly improves average sensitivity and F1-scores over FedAvg by adapting to unique patient data.
- Conducted a detailed hyperparameter analysis of the pFedMe personalization weight (λ), revealing that smaller λ values are critical for minority class detection (sensitivity), whereas larger values cause performance to collapse by forcing adherence to a globally biased model.

Optimized Convolutional Layer Implementation Using CUDA

Course Project (Advisor: Prof. Volodymyr Kindratenko)

October 2024 - December 2024

Keywords: CUDA, Parallel programming, GPU acceleration, Convolutional neural networks (CNNs)

- Designed and implemented the forward pass of convolutional layers for a modified LeNet-5 architecture using CUDA, optimizing performance for deep learning tasks such as image classification and object detection.
- Implemented a GPU-based forward convolution with a structured Prolog-Kernel-Epilog approach, ensuring memory management, convolution computation, and output transfer, while matching CPU implementation correctness and optimizing performance using Nsight profiling tools.
- Applied advanced GPU programming techniques to optimize the implementation, including streams, GEMM kernels, and kernel fusion, to achieve a target inference time of ≤ 80 ms for 10,000 images from the Fashion MNIST dataset.

RRAM-based Heterogeneous Processing for Multimodal Brain-Computer Interfaces

The University of Hong Kong (Advisor: Dr. Zhengwu Liu)

May 2024 - July 2024

Keywords: RRAM, Multimodal brain-computer interface, EEG, Compute-in-memory, FDT

- Simulated representative heterogeneous processing paradigm of P300 signal recognition in Python by using resistive random-access memory (RRAM) with and without fixed parameter disturbance training (FDT).

- Contributed to combining the BCIC IV IIa and the P300 RSVP datasets and designing an RRAM-based multimodal recognizer that integrates components of the pre-trained EEGNet, CSP (Common Spatial Pattern), and a modality-fused classifier to create the multimodal settings.
- Achieved 2.83% higher accuracy using multimodal BCI with FDT than that without FDT and significantly outperformed the MI-alone and P300-alone results by 8.19% and 13.20%, respectively.

Raspberry Pi Based IoT System as a Private Chatbot

Course Project (Advisor: Prof. Deming Chen)

March 2024 - May 2024

Keywords: *IoT System, Raspberry Pi, Machine Learning, Deep Learning*

- Developed an IoT system using Raspberry Pi 4 as a private chatbot with face detection and speaker recognition to guarantee privacy and personal conversations as well as interactions.
- Implemented MTCNN with ResNet and dlib-based face recognition, achieving better performance with the latter; trained the system with one hundred face images for live recognition.
- Built a custom residual neural network with Keras for speaker recognition, achieving 96% accuracy.
- Integrated a server-client architecture using Google Cloud for accelerated processing and implemented speech recognition and TTS for user interaction.

Multiplayer Action Game on FPGA: Crazy Arcade

Course Project (Advisor: Prof. Zuofu Cheng)

March 2024 - May 2024

Keywords: *SystemVerilog, FPGAs, System-on-a-chip, MicroBlaze CPU, VGA*

- Used FPGA for real-time operations, integrating MicroBlaze CPU for game logic and keyboard input processing.
- Developed various modules in SystemVerilog to manage player movements, bomb mechanics, life counts, and game states, interacting through a system bus.
- Incorporated background music by PWM for sound generation, featuring distinct tracks for different game stages.

Cheat-Machine for Game 2048

Course Project (Advisor: Prof. Thomas Moon)

March 2024 - May 2024

Keywords: *Embedded DSP, Real-time Signal Processing, Image Processing, Android Studio*

- Developed an app in Android Studio to analyze a live game of 2048, recognizing board digits using image processing.
- Employed efficient template matching for multi-digit recognition, using grayscale conversion, Canny edge detection, and perspective transformation to preprocess images. Used python packages to evaluate the workflow of the application.
- Built an AI engine with an Expecti-max Search algorithm to recommend the optimal move, focusing on corner placement strategies.
- Achieved high accuracy in digit recognition (100% when properly aligned) and consistent AI performance, reaching 1024 tile in 75% of simulations.

Hook&Hair Structure 3D-Printing based on Path Control and 4D Printing Experiment Exploration

Zhejiang University (Advisor: Prof. Guanyun Wang)

June 2023 - August 2023

Keywords: *3D printing, Grasshopper, Rhino, FDM, Path-planning, 4D printing*

- Developed 3D printing techniques for complex hook and hair structures using Rhino and Grasshopper for path planning, generating G-codes for customized printing paths instead of traditional FDM (Fused Deposition Modeling) methods.
- Accomplished applications including hooked ball-mitten toys and hairy objects, requiring precise path control to avoid defects and achieve intricate designs.
- Conducted experiments in 4D printing, modeling deformable planar objects in Fusion360 that transform into stereoscopic shapes when heated.

Over-the-Air-Computation Based Federated Learning Model Establishment & Simulation

Zhejiang University (Advisor: Prof. Howard Yang)

April 2022 - April 2023

Keywords: *Edge Computing, Federated Learning, OFDM, Over-the-air Computing, Simulink, Machine Learning, Neural Networks*

- Explored an innovative approach to utilizing private data from distributed databases to train shared models, ensuring user privacy while making use of the data.
- Set up an over-the-air-computation-based communication model in Simulink which could transmit and receive massive data gradients between federated users effectively.
- Combined Machine Learning models like Linear Regression and Deep Learning models like neural networks in MATLAB codes with communication models in Simulink to implement effective edge-computing models.
- Improved the model to adapt to the Large-scale applications by exploiting and modifying existing OFDM Communication Systems.

HONORS

Outstanding Graduate, University Award	<i>Zhejiang University</i>	June 2025
Dean's List (Top 20%)	<i>UIUC</i>	Fall 2023
Scholarship from Zhejiang University (Top 8%)	<i>Zhejiang University</i>	2021-2022, 2022-2023 and 2023-2024 Academic Years
Academic Excellence Award	<i>Zhejiang University</i>	2021-2022, 2022-2023 and 2023-2024 Academic Years
Outstanding Performance in Social Work Award	<i>Zhejiang University</i>	2021-2022 Academic Year
Recognition for Outstanding Performance, 2022 Undergraduate Summer Research Program	<i>Zhejiang University</i>	Summer 2022
Recognition for Academic Poster Exhibition, 2022 Undergraduate Summer Research Program	<i>Zhejiang University</i>	Summer 2022

SKILLS

Programming Languages: Python, C/C++, System Verilog, CUDA, MATLAB, LC-3 assembly
Softwares: Visual Studio Code, Jupyter Notebook, MATLAB, Rhino, Android Studio
Hardwares: FPGA, STM32 MCU, 3D Printing, PCB design